

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

Chapter 2

With Question/Answer Animations

Chapter Summary

- Sets
 - The Language of Sets
 - Set Operations
 - Set Identities
- Functions
 - Types of Functions
 - Operations on Functions
 - Computability
- Sequences and Summations
 - Types of Sequences
 - Summation Formulae

Sets

Section 2.1

Section Summary

- Definition of sets
- Describing Sets
 - Roster Method
 - Set-Builder Notation
- Some Important Sets in Mathematics
- Empty Set and Universal Set
- Subsets and Set Equality
- Cardinality of Sets
- Tuples
- Cartesian Product

Introduction

- Sets are one of the basic building blocks for the types of objects considered in discrete mathematics.
 - Important for counting.
 - Programming languages have set operations.
- Set theory is an important branch of mathematics.
 - Many different systems of axioms have been used to develop set theory.
 - Here we are not concerned with a formal set of axioms for set theory. Instead, we will use what is called naïve set theory.

Sets

- A *set* is an unordered collection of objects.
 - the students in this class
 - the chairs in this room
- The objects in a set are called the *elements*, or *members* of the set. A set is said to *contain* its elements.
- The notation $a \in A$ denotes that a is an element of the set A .
- If a is not a member of A , write $a \notin A$

Describing a Set: Roster Method

- $S = \{a, b, c, d\}$
- Order not important

$$S = \{a, b, c, d\} = \{b, c, a, d\}$$

- Each distinct object is either a member or not; listing more than once does not change the set.

$$S = \{a, b, c, d\} = \{a, b, c, b, c, d\}$$

- Ellipses (...) may be used to describe a set without listing all of the members when the pattern is clear.

$$S = \{a, b, c, d, \dots, z\}$$

Roster Method

- Set of all vowels in the English alphabet:

$$V = \{a, e, i, o, u\}$$

- Set of all odd positive integers less than 10:

$$O = \{1, 3, 5, 7, 9\}$$

- Set of all positive integers less than 100:

$$S = \{1, 2, 3, \dots, 99\}$$

- Set of all integers less than 0:

$$S = \{\dots, -3, -2, -1\}$$

Some Important Sets

$\mathbf{N}$ = *natural numbers* = $\{0,1,2,3,\dots\}$

$\mathbf{Z}$ = *integers* = $\{\dots,-3,-2,-1,0,1,2,3,\dots\}$

$\mathbf{Z}^+$ = *positive integers* = $\{1,2,3,\dots\}$

$\mathbf{R}$ = *set of real numbers*

$\mathbf{R}^+$ = *set of positive real numbers*

$\mathbf{C}$ = *set of complex numbers.*

$\mathbf{Q}$ = *set of rational numbers*

Set-Builder Notation

- Specify the property or properties that all members must satisfy:

$$S = \{x \mid x \text{ is a positive integer less than } 100\}$$

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

$$O = \{x \in \mathbf{Z}^+ \mid x \text{ is odd and } x < 10\}$$

- A predicate may be used:

$$S = \{x \mid P(x)\}$$

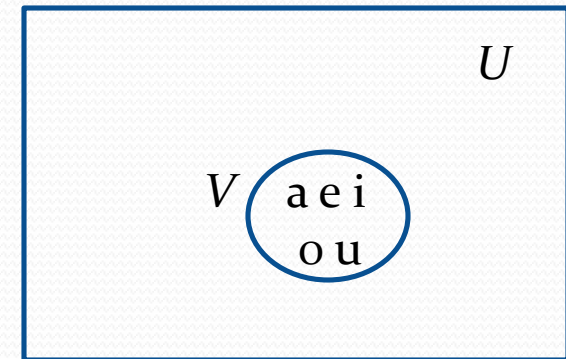
- Example: $S = \{x \mid \text{Prime}(x)\}$
- Positive rational numbers:

$$\mathbf{Q}^+ = \{x \in \mathbf{R} \mid x = p/q, \text{ for some positive integers } p, q\}$$

Universal Set and Empty Set

- The *universal set* U is the set containing everything currently under consideration.
 - Sometimes implicit
 - Sometimes explicitly stated.
 - Contents depend on the context.
- The empty set is the set with no elements. Symbolized $\emptyset$, but $\{\}$ also used.

Venn Diagram



John Venn (1834-1923)
Cambridge, UK

Some things to remember

- Sets can be elements of sets.

$$\{\{1,2,3\}, a, \{b,c\}\}$$

$$\{\mathbf{N}, \mathbf{Z}, \mathbf{Q}, \mathbf{R}\}$$

- The empty set is different from a set containing the empty set.

$$\emptyset \neq \{ \emptyset \}$$

Set Definitions

The Empty Set (Null Set)

- Set that has no elements
- Denoted by $\emptyset$ or $\{\}$

Example:

The set of all positive integers that are greater than their squares is the null set.

Singleton set

- A set with one element is called a singleton set.

Example:

$$A = \{2\}$$

Set Equality

Definition: Two sets are *equal* if and only if they have the same elements.

- Therefore if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$
- We write $A = B$ if A and B are equal sets.

$$\{1, 3, 5\} = \{3, 5, 1\}$$

$$\{1, 5, 5, 5, 3, 3, 1\} = \{1, 3, 5\}$$

Set Equality

✓ Two sets are **equal** if and only if **they have the same elements**.

That is, if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$.

We write **$A = B$** if A and B are equal sets.

• $A = \{9, 2, 7, -3\}, B = \{7, 9, -3, 2\} :$ **$A = B$**

• $A = \{\text{dog}, \text{cat}, \text{horse}\},$
 $B = \{\text{cat}, \text{horse}, \text{elephant}, \text{dog}\} :$ **$A \neq B$**

• $A = \{\text{dog}, \text{cat}, \text{horse}\},$
 $B = \{\text{cat}, \text{horse}, \text{dog}, \text{dog}\} :$ **$A = B$**

Subsets

Definition: The set A is a *subset* of B , if and only if every element of A is also an element of B .

- The notation $A \subseteq B$ is used to indicate that A is a subset of the set B .
- $A \subseteq B$ holds if and only if $\forall x(x \in A \rightarrow x \in B)$ is true.
 1. Because $a \in \emptyset$ is always false, $\emptyset \subseteq S$, for every set S .
 2. Because $a \in S \rightarrow a \in S$, $S \subseteq S$, for every set S .

Subsets

- The set A is a subset of B iff every element of A is also an element of B .

$$A \subseteq B$$

- The null set is a subset of every set.

$$\emptyset \subseteq A$$

- Every set is a subset of itself.

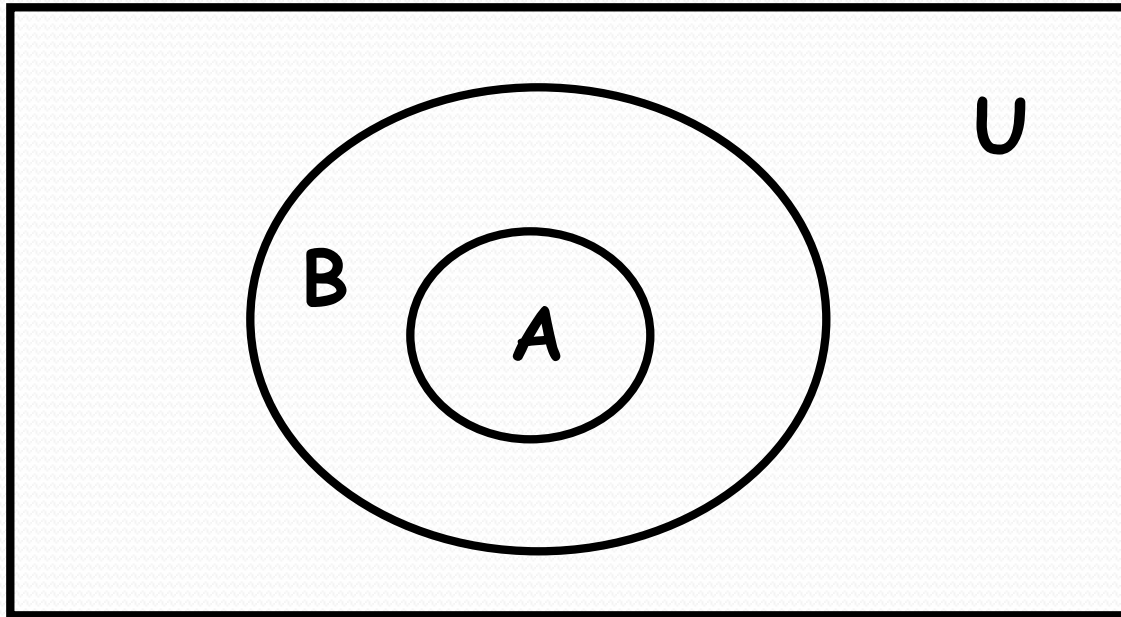
$$A \subseteq A$$

- A is a **proper** subset of B , if A is a subset of B and A is not equal to B .

$$A \subset B$$

Subsets

$A \subseteq B$ "A is a subset of the set B".



Venn Diagram Showing that
A Is a Subset of B.

Subsets

$A \subseteq B$ "A is a subset of the set B".

Examples:

$A = \{3, 9\}, B = \{5, 9, 1, 3\}$

$A \subseteq B ?$

true

$A = \{3, 3, 3, 9\}, B = \{5, 9, 1, 3\}$

$A \subseteq B ?$

true

$A = \{1, 2, 3\}, B = \{2, 3, 4\}$

$A \subseteq B ?$

false

Showing a Set is or is not a Subset of Another Set

- **Showing that A is a Subset of B:** To show that $A \subseteq B$, show that if x belongs to A , then x also belongs to B .
- **Showing that A is not a Subset of B:** To show that A is not a subset of B , $A \not\subseteq B$, find an element $x \in A$ with $x \notin B$. (Such an x is a counterexample to the claim that $x \in A$ implies $x \in B$.)

Examples:

1. The set of all computer science majors at your school is a subset of all students at your school.
2. The set of integers with squares less than 100 is not a subset of the set of nonnegative integers.

Another look at Equality of Sets

- Recall that two sets A and B are *equal*, denoted by $A = B$, iff
$$\forall x (x \in A \leftrightarrow x \in B)$$

- Using logical equivalences we have that $A = B$ iff

$$\forall x [(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$$

- This is equivalent to

$$A \subseteq B \quad \text{and} \quad B \subseteq A$$

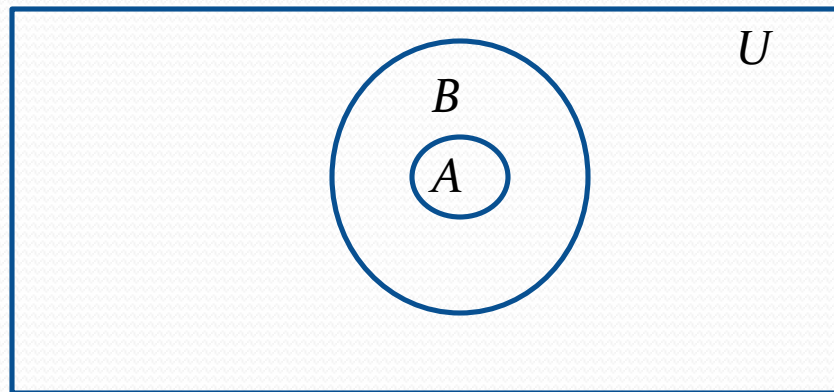
Proper Subsets

Definition: If $A \subseteq B$, but $A \neq B$, then we say A is a *proper subset* of B , denoted by $A \subset B$. If $A \subset B$, then

$$\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$$

is true.

Venn Diagram



Set Cardinality

Definition: If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is *finite*. Otherwise it is *infinite*.

Definition: The *cardinality* of a finite set A , denoted by $|A|$, is the number of (distinct) elements of A .

Examples:

1. $|\emptyset| = |\{\ \}| = 0$
2. Let S be the letters of the English alphabet. Then $|S| = 26$
3. $|\{1,2,3\}| = 3$
4. $|\{\emptyset\}| = 1$
5. The set of integers is infinite.

Power Sets

Definition: The set of all subsets of a set A , denoted $P(A)$, is called the *power set* of A .

Example: If $A = \{a, b\}$ then

$$\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

- If a set has n elements, then the cardinality of the power set is 2^n . (In Chapters 5 and 6, we will discuss different ways to show this.)

The Power Set

- ✓ What is the power set of the set $\{0, 1, 2\}$?

$$P(\{0, 1, 2\}) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}.$$

Note that the **empty set** and the **set itself** are members of this set of subsets.

- ✓ What is the power set of the empty set?

$$P(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}.$$

Tuples

- The *ordered n -tuple* $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element and a_n as its last element.
- Two n -tuples are equal if and only if their corresponding elements are equal.
- 2-tuples are called *ordered pairs*.
- The ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$.

Cartesian Product



René Descartes
(1596-1650)

Definition: The *Cartesian Product* of two sets A and B , denoted by $A \times B$ is the set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) | a \in A \wedge b \in B\}$$

Example:

$$A = \{a, b\} \quad B = \{1, 2, 3\}$$

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

$$A \times B \neq B \times A$$

- **Definition:** A subset R of the Cartesian product $A \times B$ is called a *relation* from the set A to the set B . (Relations will be covered in depth in Chapter 9.)

Cartesian Products

✓ The Cartesian products $A \times B$ and $B \times A$ are not equal, unless

$A = \emptyset$ or $B = \emptyset$ (so that $A \times B = \emptyset$)

or

$A = B$

Cartesian Product

Definition: The cartesian products of the sets $A_1, A_2, \dots, A_n$, denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of ordered n -tuples $(a_1, a_2, \dots, a_n)$ where a_i belongs to A_i for $i = 1, \dots, n$.

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$$

Example: What is $A \times B \times C$ where $A = \{0,1\}$, $B = \{1,2\}$ and $C = \{0,1,2\}$

Solution: $A \times B \times C = \{(0,1,0), (0,1,1), (0,1,2), (0,2,0), (0,2,1), (0,2,2), (1,1,0), (1,1,1), (1,1,2), (1,2,0), (1,2,1), (1,2,2)\}$

Set Operations

Section 2.2

Section Summary

- Set Operations
 - Union
 - Intersection
 - Complementation
 - Difference
- More on Set Cardinality
- Set Identities
- Proving Identities
- Membership Tables

Union

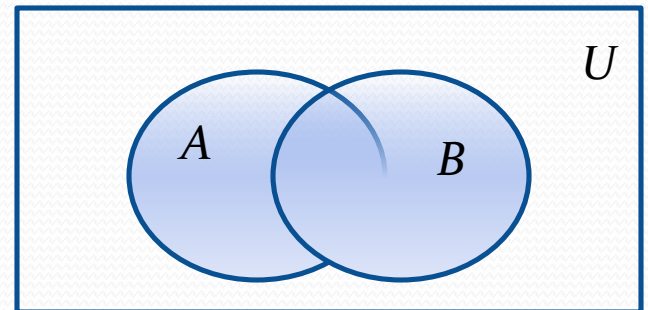
- **Definition:** Let A and B be sets. The *union* of the sets A and B , denoted by $A \cup B$, is the set:

$$\{x | x \in A \vee x \in B\}$$

- **Example:** What is $\{1,2,3\} \cup \{3,4,5\}$?

Solution: $\{1,2,3,4,5\}$

Venn Diagram for $A \cup B$



Intersection

- **Definition:** The *intersection* of sets A and B , denoted by $A \cap B$, is

$$\{x | x \in A \wedge x \in B\}$$

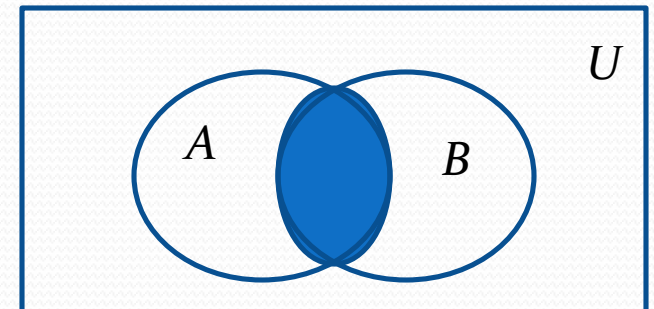
- Note if the intersection is empty, then A and B are said to be *disjoint*.
- **Example:** What is? $\{1,2,3\} \cap \{3,4,5\}$?

Solution: $\{3\}$

- **Example:** What is?
 $\{1,2,3\} \cap \{4,5,6\}$?

Solution: $\emptyset$

Venn Diagram for $A \cap B$



Complement

Definition: If A is a set, then the complement of the A (with respect to U), denoted by $\bar{A}$ is the set $U - A$

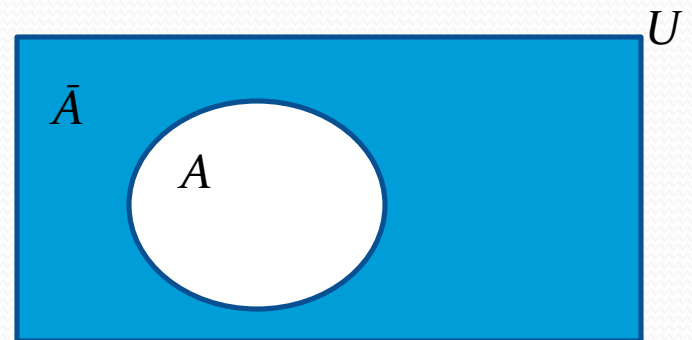
$$\bar{A} = \{x \in U \mid x \notin A\}$$

(The complement of A is sometimes denoted by A^c .)

Example: If U is the positive integers less than 100, what is the complement of $\{x \mid x > 70\}$

Solution: $\{x \mid x \leq 70\}$

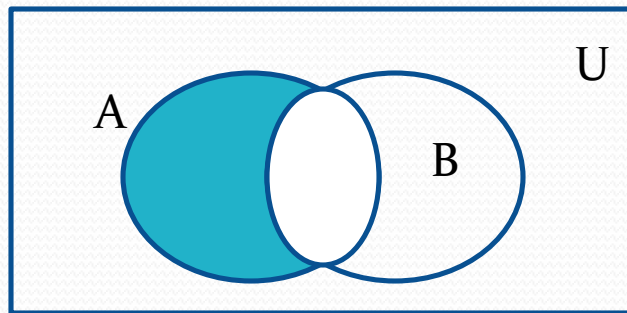
Venn Diagram for Complement



Difference

- **Definition:** Let A and B be sets. The *difference* of A and B , denoted by $A - B$, is the set containing the elements of A that are not in B . The difference of A and B is also called the complement of B with respect to A .

$$A - B = \{x \mid x \in A \wedge x \notin B\} = A \cap \bar{B}$$

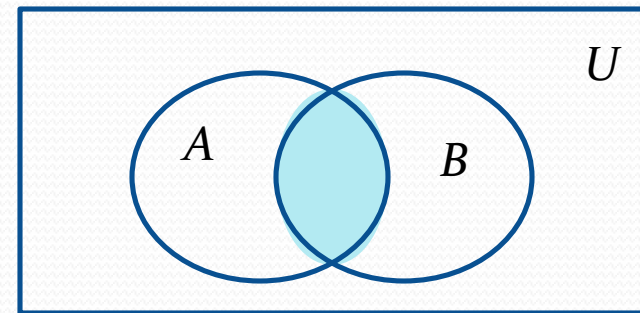


Venn Diagram for $A - B$

The Cardinality of the Union of Two Sets

- Inclusion-Exclusion

$$|A \cup B| = |A| + |B| - |A \cap B|$$



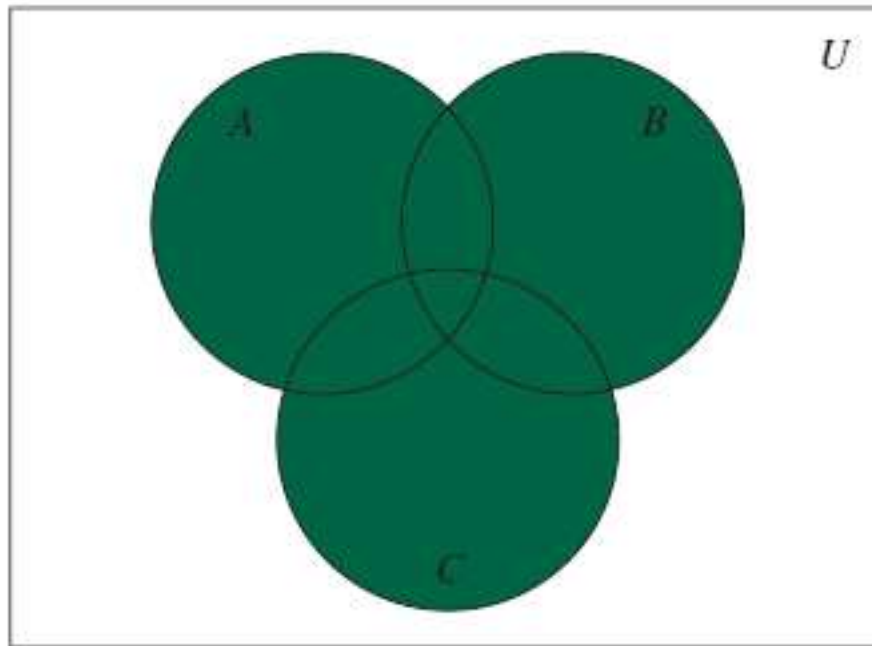
Venn Diagram for A , B , $A \cap B$, $A \cup B$

- **Example:** Let A be the math majors in your class and B be the CS majors. To count the number of students who are either math majors or CS majors, add the number of math majors and the number of CS majors, and subtract the number of joint CS/math majors.
- We will return to this principle in Chapter 6 and Chapter 8 where we will derive a formula for the cardinality of the union of n sets, where n is a positive integer.

Generalized Unions and Intersections

✓ $A \cup B \cup C$

The set contains those elements that are in at least one of the sets A , B , and C .

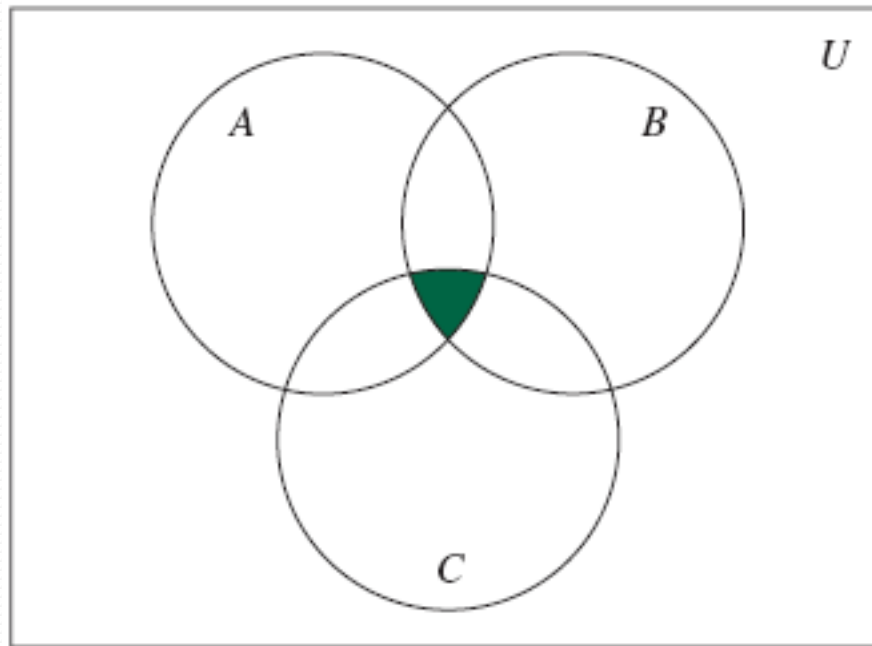


$A \cup B \cup C$ is shaded

Generalized Unions and Intersections

✓ $A \cap B \cap C$

The set contains those elements that are in all of A , B , and C .



$A \cap B \cap C$ is shaded

Generalized Unions and Intersections

Example

Let $A = \{0, 2, 4, 6, 8\}$, $B = \{0, 1, 2, 3, 4\}$, and $C = \{0, 3, 6, 9\}$. What are $A \cup B \cup C$ and $A \cap B \cap C$?

The set $A \cup B \cup C$ contains those elements in at least one of A , B , and C . Hence,

$$A \cup B \cup C = \{0, 1, 2, 3, 4, 6, 8, 9\}.$$

The set $A \cap B \cap C$ contains those elements in all three of A , B , and C . Thus,

$$A \cap B \cap C = \{0\}.$$

Review Questions

Example: $U = \{0,1,2,3,4,5,6,7,8,9,10\}$ $A = \{1,2,3,4,5\}$, $B = \{4,5,6,7,8\}$

1. $A \cup B$

Solution: $\{1,2,3,4,5,6,7,8\}$

2. $A \cap B$

Solution: $\{4,5\}$

3. $\bar{A}$

Solution: $\{0,6,7,8,9,10\}$

4. $\bar{B}$

Solution: $\{0,1,2,3,9,10\}$

5. $A - B$

Solution: $\{1,2,3\}$

6. $B - A$

Solution: $\{6,7,8\}$

Symmetric Difference (*optional*)

Definition: The *symmetric difference* of **A** and **B**, denoted by $A \oplus B$ is the set

$$(A - B) \cup (B - A)$$

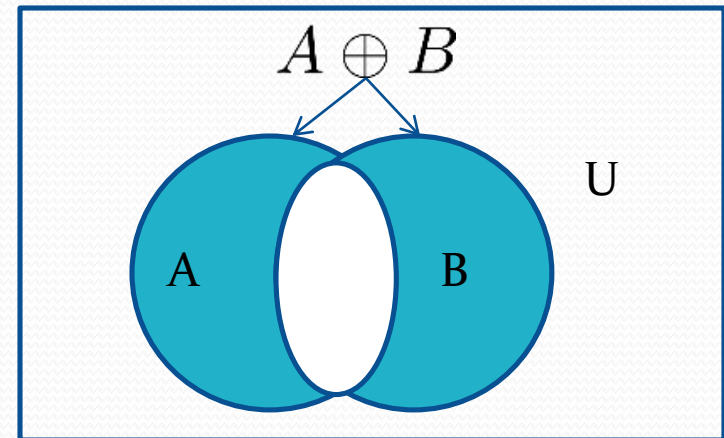
Example:

$$U = \{0,1,2,3,4,5,6,7,8,9,10\}$$

$$A = \{1,2,3,4,5\} \quad B = \{4,5,6,7,8\}$$

What is:

- **Solution:** $\{1,2,3,6,7,8\}$



Venn Diagram

Set Identities

- Identity laws

$$A \cup \emptyset = A \quad A \cap U = A$$

- Domination laws

$$A \cup U = U \quad A \cap \emptyset = \emptyset$$

- Idempotent laws

$$A \cup A = A \quad A \cap A = A$$

- Complementation law

$$\overline{(\overline{A})} = A$$

Continued on next slide →

Set Identities

- Commutative laws

$$A \cup B = B \cup A \quad A \cap B = B \cap A$$

- Associative laws

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

- Distributive laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Continued on next slide →

Set Identities

- De Morgan's laws

$$\overline{A \cup B} = \overline{A} \cap \overline{B} \quad \overline{A \cap B} = \overline{A} \cup \overline{B}$$

- Absorption laws

$$A \cup (A \cap B) = A \quad A \cap (A \cup B) = A$$

- Complement laws

$$A \cup \overline{A} = U \quad A \cap \overline{A} = \emptyset$$

Proving Set Identities

- Different ways to prove set identities:
 1. Prove that each set (side of the identity) is a subset of the other.
 2. Use set builder notation and propositional logic.
 3. Membership Tables: Verify that elements in the same combination of sets always either belong or do not belong to the same side of the identity. Use 1 to indicate it is in the set and a 0 to indicate that it is not.

Set Identities

✓ prove that $\overline{A \cup (B \cap C)} = (\overline{C} \cup \overline{B}) \cap \overline{A}$

$$\begin{aligned}\overline{A \cup (B \cap C)} &= \overline{A} \cap \overline{(B \cap C)} \\ &= \overline{A} \cap (\overline{B} \cup \overline{C}) \\ &= (\overline{B} \cup \overline{C}) \cap \overline{A} \\ &= (\overline{C} \cup \overline{B}) \cap \overline{A}\end{aligned}$$

Proof of Second De Morgan Law

Example: Prove that $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Solution: We prove this identity by showing that:

$$1) \quad \overline{A \cap B} \subseteq \overline{A} \cup \overline{B} \quad \text{and}$$

$$2) \quad \overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$$

Continued on next slide →

Proof of Second De Morgan Law

These steps show that: $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$

$$x \in \overline{A \cap B}$$

by assumption

$$x \notin A \cap B$$

defn. of complement

$$\neg((x \in A) \wedge (x \in B))$$

defn. of intersection

$$\neg(x \in A) \vee \neg(x \in B)$$

1st De Morgan Law for Prop Logic

$$x \notin A \vee x \notin B$$

defn. of negation

$$x \in \overline{A} \vee x \in \overline{B}$$

defn. of complement

$$x \in \overline{A} \cup \overline{B}$$

defn. of union

Continued on next slide →

Proof of Second De Morgan Law

These steps show that:

$$\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$$

$$x \in \overline{A} \cup \overline{B}$$

by assumption

$$(x \in \overline{A}) \vee (x \in \overline{B})$$

defn. of union

$$(x \notin A) \vee (x \notin B)$$

defn. of complement

$$\neg(x \in A) \vee \neg(x \in B)$$

defn. of negation

$$\neg((x \in A) \wedge (x \in B))$$

by 1st De Morgan Law for Prop Logic

$$\neg(x \in A \cap B)$$

defn. of intersection

$$x \in \overline{A \cap B}$$

defn. of complement



Set-Builder Notation: Second De Morgan Law

$\overline{A \cap B}$	$=$	$\{x x \notin A \cap B\}$	by defn. of complement
	$=$	$\{x \neg(x \in (A \cap B))\}$	by defn. of does not belong symbol
	$=$	$\{x \neg(x \in A \wedge x \in B)\}$	by defn. of intersection
	$=$	$\{x \neg(x \in A) \vee \neg(x \in B)\}$	by 1st De Morgan law for Prop Logic
	$=$	$\{x x \notin A \vee x \notin B\}$	by defn. of not belong symbol
	$=$	$\{x x \in \overline{A} \vee x \in \overline{B}\}$	by defn. of complement
	$=$	$\{x x \in \overline{A} \cup \overline{B}\}$	by defn. of union
	$=$	$\overline{A} \cup \overline{B}$	by meaning of notation



Membership Table

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

A	B	C	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0